Exercise 1. Suppose $x, y \in \mathbb{R}$ and $x < y$. Prove that there exists $i \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < i < y$.

Proof. Let $x, y \in \mathbb{R}$ with $x < y$. By the Density of $\mathbb{Q}$, $\exists r \in \mathbb{Q}$ such that

$$\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}} \iff x < \sqrt{2}r < y.$$

Suppose, for example, $i = \sqrt{2}r$. Assume, for the purpose of contradiction, $i \in \mathbb{Q}$. Then

$$\sqrt{2}r \in \mathbb{Q} \text{ (and } r \in \mathbb{Q}) \implies \sqrt{2} \in \mathbb{Q} \text{ (because } r^{-1} \in \mathbb{Q}),$$

which is a contradiction. Thus, $i \in \mathbb{R} \setminus \mathbb{Q}$ and $x < i < y$, as needed. □

Exercise 2. Let $E \subset (0, 1)$ be the set of all real numbers with decimal representation using only the digits 1 and 2:

$$E := \{x \in (0, 1) : \forall j \in \mathbb{N},\ \exists d_{-j} \in \{1, 2\} \text{ such that } x = 0.d_{-1}d_{-2}...\}.$$

Prove that $|E| = |P(\mathbb{N})|$. *Hint*: Consider the function $f : E \to P(\mathbb{N})$ such that if $x \in E$ and $x = 0.d_{-1}d_{-2}...$,

$$f(x) = \{j \in \mathbb{N} : d_{-j} = 2\}.$$

Proof. Let $E := \{x \in (0, 1) : \forall j \in \mathbb{N}\ \exists d_{-j} \in \{1, 2\}\ x = 0.d_{-1}d_{-2}...\} \subset \mathbb{R}$. We want to prove that $|E| = |P(\mathbb{N})|$. Consider $f : E \to P(\mathbb{N})$ such that $f(x) = \{j \in \mathbb{N} : d_{-j} = 2\} \subset \mathbb{N}$ if $x \in E$ and $x = 0.d_{-1}d_{-2}...$ We need to prove that $f : E \to P(\mathbb{N})$ is injective. Assume $f(x) = f(y)$. Without loss of generality, let $x = 0.d_{-1}d_{-2}...$ and $y = 0.d'_{-1}d'_{-2}...$ Then

$$\forall j \in \mathbb{N} : d_{-j} = d'_{-j} \implies \forall j \in \mathbb{N} : 0.d_{-1}d_{-2}...d_{-j} = 0.d'_{-1}d'_{-2}...d'_{-j};$$

hence,

$$x = \sup\{0.d_{-1}d_{-2}...d_{-j} : j \in \mathbb{N}\} = \sup\{0.d'_{-1}d'_{-2}...d'_{-j} : j \in \mathbb{N}\} = y,$$

as needed. We need to prove that $f : E \to P(\mathbb{N})$ is surjective. Assume $N \in P(\mathbb{N})$. By construction, $\exists x \in E$ such that $x = 0.d_{-1}d_{-2}...$ with $\forall m \in \mathbb{N} \setminus N : d_{-m} = 1$ and $\forall n \in N : d_{-n} = 2$. Then

$$N = \{j \in \mathbb{N} : d_{-j} = 2\} = f(x) \in f(E),$$

as needed. Thus, $f : E \to P(\mathbb{N})$ is bijective, as wanted. □

Exercise 3. Prove:

(a) Let A and B be two disjoint, countably infinite sets. Then $A \cup B$ is countably infinite.

(b) The set of irrational numbers, $\mathbb{R} \setminus \mathbb{Q}$, is uncountable. You may use the facts discussed in the lectures that $\mathbb{R} \setminus \mathbb{Q}$ is infinite and $\mathbb{R}$ is uncountable without proof.

Proof. Start:

(a) Let A and B be two disjoint, countably infinite sets. Then $\exists$ bijective functions $f : A \to \mathbb{N}$ and $g : B \to \mathbb{N}$. Define the function $h : A \cup B \to \mathbb{N}$ such that

$$h(x) = \begin{cases} 2f(x) - 1 & (x \in A) \\ 2g(x) & (x \in B) \end{cases}.$$

We need to prove that h is injective. Assume $h(x) = h(y)$. Then

$$\begin{aligned} h(x) \in \{2n - 1 : n \in \mathbb{N}\} &\implies 2f(x) - 1 = 2f(y) - 1 \\ &\implies f(x) = f(y) \\ &\implies x = y \text{ (because } f \text{ is injective)} \end{aligned}$$

xor

$$\begin{aligned} h(y) \in \{2n : n \in \mathbb{N}\} &\implies 2g(x) = 2g(y) \\ &\implies g(x) = g(y) \\ &\implies x = y \text{ (because } g \text{ is injective);} \end{aligned}$$

hence, $x = y \in A \cup B$, as needed. We need to prove that h is surjective. Assume $n \in \mathbb{N}$. Then

$$\begin{aligned} n \in \{2m - 1 : m \in \mathbb{N}\} &\implies \exists x \in A : f(x) = \frac{n+1}{2} \in \mathbb{N} \text{ (because } f \text{ is surjective)} \\ &\implies h(x) = 2f(x) - 1 = n \implies n \in h(A \cup B) \end{aligned}$$

xor

$$\begin{aligned} n \in \{2m : m \in \mathbb{N}\} &\implies \exists y \in B : g(y) = \frac{n}{2} \in \mathbb{N} \text{ (because } g \text{ is surjective)} \\ &\implies h(y) = 2g(y) = n \implies n \in h(A \cup B); \end{aligned}$$

hence, $n \in h(A \cup B)$, as needed. Thus, $h : A \cup B \to \mathbb{N}$ is bijective; hence, $A \cup B$ is countably infinite, as needed.

(b) Assume, for the purpose of contradiction, $\mathbb{R} \setminus \mathbb{Q}$ is countable. Given $\mathbb{R} \setminus \mathbb{Q}$ is infinite, $\mathbb{R} \setminus \mathbb{Q}$ is countably infinite. Note that $\mathbb{Q}$ is countably infinite. By Part (a),

$$|\mathbb{R} \setminus \mathbb{Q}| = |\mathbb{N}| = |\mathbb{Q}| \implies |\mathbb{R}| = |\mathbb{R} \setminus \mathbb{Q} \cup \mathbb{Q}| = |\mathbb{N}|;$$

hence, $\mathbb{R}$ is countably infinite, which is a contradiction. Thus, $\mathbb{R} \setminus \mathbb{Q}$ is uncountable, as needed.

□

Exercise 4. Let S be a subset of $\mathbb{R}$ which is bounded above, and let x_0 be an upper bound for S. Prove that $x_0 = \sup S$ if and only if for every $\epsilon > 0$, there exists $x \in S$ such that $x_0 - \epsilon < x$.

Proof. Let S be any bounded above subset of $\mathbb{R}$. Thus, $\exists x_0 \in \mathbb{R}$ such that x_0 is an upper bound for S, as wanted. If x_0 is the least upper bound for S, then

$$\forall \epsilon > 0 : x_0 - \epsilon < x_0 \implies \exists x \in S : x_0 - \epsilon < x \leq x_0,$$

as needed. If $\forall \epsilon > 0 \; \exists x \in S : x_0 - \epsilon < x \leq x_0$, then

$$\forall x \in \mathbb{R} : x < x_0 \implies \epsilon_0 = x_0 - x > 0 \implies \exists y \in S : x = x_0 - \epsilon_0 < y \leq x_0,$$

as needed. Thus, $x_0 = \sup S$ if and only if $\forall \epsilon > 0 \; \exists x \in S : x_0 - \epsilon < x \leq x_0$, as wanted. □

Exercise 5. We say a set $U \subset \mathbb{R}$ is *open* if for every $x \in U$ there exists $\epsilon > 0$ such that

$$(x - \epsilon, x + \epsilon) \subset U.$$

Since the definition is vacuous for $U = \emptyset$, it follows that the empty set is open. It is also clear from the definition that $U = \mathbb{R}$ is open.

(a) Let $a, b \in \mathbb{R}$ with $a < b$. Prove that the sets $(-\infty, a)$, (a, b), and (b, ∞) are open.

(b) Let Λ be a set (not necessarily a subset of $\mathbb{R}$), and for each $\lambda \in \Lambda$, let $U_\lambda \subset \mathbb{R}$. Prove that if U_λ is open for all $\lambda \in \Lambda$ then the set

$$\bigcup_{\lambda \in \Lambda} U_\lambda = \{x \in \mathbb{R} : \exists \lambda \in \Lambda \text{ such that } x \in U_\lambda\}$$

is open.

(c) Let $n \in \mathbb{N}$, and let $U_1, ..., U_n \subset \mathbb{R}$. Prove that if $U_1, ..., U_n$ are open then the set

$$\bigcap_{m=1}^{n} U_m = \{x \in \mathbb{R} : x \in U_m \text{ for all } m = 1, ..., n\}$$

is open.

(d) Is the set of rationals $\mathbb{Q} \subset \mathbb{R}$ open? Provide a proof to substantiate your claim.

Proof. Start!

(a) Let $a, b \in \mathbb{R}$ with $a < b$. We need to prove that the subset $(-\infty, a)$ is open. If $x \in (-\infty, a)$, then

$$-\infty < x < a \implies \epsilon_0 = \frac{a - x}{2} > 0 \implies -\infty < x - \epsilon_0 < x + \epsilon_0 < a;$$

hence, $\exists \epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subset (-\infty, a)$, as needed. We need to prove that the subset (a, b) is open. If $x \in (a, b)$, then

$$a < x < b \implies \epsilon_0 = \min\left\{\frac{x - a}{2}, \frac{b - x}{2}\right\} > 0 \implies a < x - \epsilon_0 < x + \epsilon_0 < b;$$

hence, $\exists \epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subset (a, b)$, as needed. We need to prove that the subset (b, ∞) is open. If $x \in (b, \infty)$, then

$$b < x < \infty \implies \epsilon_0 = \frac{x - b}{2} > 0 \implies b < x - \epsilon_0 < x + \epsilon_0 < \infty;$$

hence, $\exists \epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \subset (b, \infty)$, as needed.

(b) Let Λ be any index set and $U_\lambda \subset \mathbb{R}$ for each $\lambda \in \Lambda$. Assume U_λ is open for all $\lambda \in \Lambda$. We want to prove that the subset

$$\bigcup_{\lambda\in\Lambda} U_\lambda = \{x \in \mathbb{R} : \exists \lambda \in \Lambda \text{ such that } x \in U_\lambda\}$$

is open. Thus,

$$\begin{aligned} x \in \bigcup_{\lambda\in\Lambda} U_\lambda &\implies \exists \lambda \in \Lambda : x \in U_\lambda \\ &\implies \exists \epsilon > 0 : (x-\epsilon, x+\epsilon) \subset U_\lambda \implies (x-\epsilon, x+\epsilon) \subset \bigcup_{\lambda\in\Lambda} U_\lambda, \end{aligned}$$

as wanted.

(c) Let $n \in \mathbb{N}$ and $U_1, ..., U_n \subset \mathbb{R}$. Assume $U_1, ..., U_n$ are open. We want to prove that

$$\bigcap_{m=1}^{n} U_m = \{x \in \mathbb{R} : x \in U_m \text{ for all } m = 1, ..., n\}$$

is open. Thus,

$$\begin{aligned} x \in \bigcap_{m=1}^{n} U_m &\implies \forall m \in \{1, ..., n\} : x \in U_m \\ &\implies \exists \epsilon_m > 0 : (x-\epsilon_m, x+\epsilon_m) \subset U_m \\ &\implies \epsilon = \min\{\epsilon_1, ..., \epsilon_n\} > 0 \implies (x-\epsilon, x+\epsilon) \subset \bigcap_{m=1}^{n} U_m, \end{aligned}$$

as wanted.

(d) No, I claim the set of rationals $\mathbb{Q}$ cannot be open in the set of reals $\mathbb{R}$. If $x \in \mathbb{Q}$, then

$$\forall \epsilon > 0 \ \exists i \in \mathbb{R} \setminus \mathbb{Q} : i \in (x-\epsilon, x+\epsilon) \implies (x-\epsilon, x+\epsilon) \not\subset \mathbb{Q};$$

hence, x is not an interior point of $\mathbb{Q}$, as needed. Thus, $\mathbb{Q}$ is not an open subset of $\mathbb{R}$, as claimed.

□

Exercise 6. Prove that

$$\lim_{n\to\infty} \frac{1}{20n^2 + 20n + 2020} = 0.$$

Proof. Let $\epsilon > 0$. By the Archimedean Property of $\mathbb{R}$, $\exists M \in \mathbb{N}$ such that $M\epsilon > 1$. Then $\forall n \geq M \geq 1$,

$$\left|\frac{1}{20n^2 + 20n + 2020} - 0\right| = \frac{1}{20n^2 + 20n + 2020} < \frac{1}{20n^2 + 20n} < \frac{1}{20n} < \frac{1}{n} \leq \frac{1}{M} < \epsilon,$$

as needed. □